

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Data Interpretation

Weightage: 20%

QUESTIONS:

Total Marks: 20

1: HTTP Response Analysis

A web server returns the following HTTP response header:

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive
```

Questions:

1. Identify any incorrect or invalid fields in the header.
2. Analyze the issue with `Content-Length` and explain its impact.
3. Determine whether the status code matches the response correctness.
4. Suggest corrections for the identified errors.

2: DOM Structure Analysis

Consider the following DOM tree representation:

```
<html>
  <head>
    <title>Test Page</title>
  </head>
  <body>
    <div>
      <p>Hello World
    </div>
  </body>
</html>
```

Questions:

1. Identify missing or improperly closed elements.
2. Analyze how the browser will interpret this structure.
3. Determine the impact on rendering and layout.
4. Suggest corrections to make the DOM valid.

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in body	Moderate

Questions:

1. Interpret the differences in data transmission between GET and POST.
2. Analyze which method is more suitable for login and why.
3. Identify security risks in using GET for authentication.
4. Suggest best practices for secure login implementation.

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Sample</title>
  </head>
  <body>
    <h1>Welcome
    <p>This is a paragraph
    
  </body>
</html>
```

Questions:

1. Identify missing closing tags.
2. Analyze issues with the `<img>` tag.
3. Determine how browsers will handle these errors.
4. Suggest a corrected version of the code.

5: Browser-Server Communication Flow

The following sequence describes a web request:

1. User enters a URL in the browser
2. DNS lookup occurs
3. HTTP request is sent
4. Server processes request
5. Response is returned
6. Browser renders page

Questions:

1. Interpret each step in the communication flow.
2. Identify where delays can occur.
3. Analyze the role of DNS in the process.
4. Suggest optimizations to improve performance.

Code of Conduct:

I ____K.Deepika(192371042)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:k.deepika

Optimized for Speed, Built for Standards

A demonstration of modern browser-server delivery mechanisms,
clean DOM structure, and responsive layout performance.

[Explore Architecture](#)

DNS & Resource Hints

Pre-resolving domain names and pre-connecting to critical origins minimizes TCP handshake delays before assets are explicitly called.

DOM Structure

Strictly compliant HTML5 tree structures ensure smooth parser processing, predictable CSS styling, and zero layout bugs.

Asset Optimization

Utilizing high-priority fetching for key assets while enforcing native lazy-loading for non-critical elements reduces initial payload size.

DNS & Resource Hints

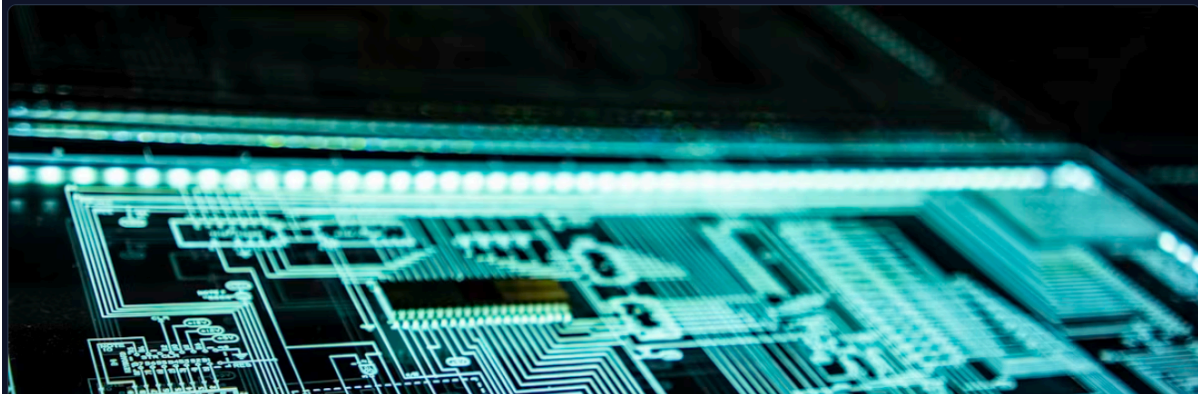
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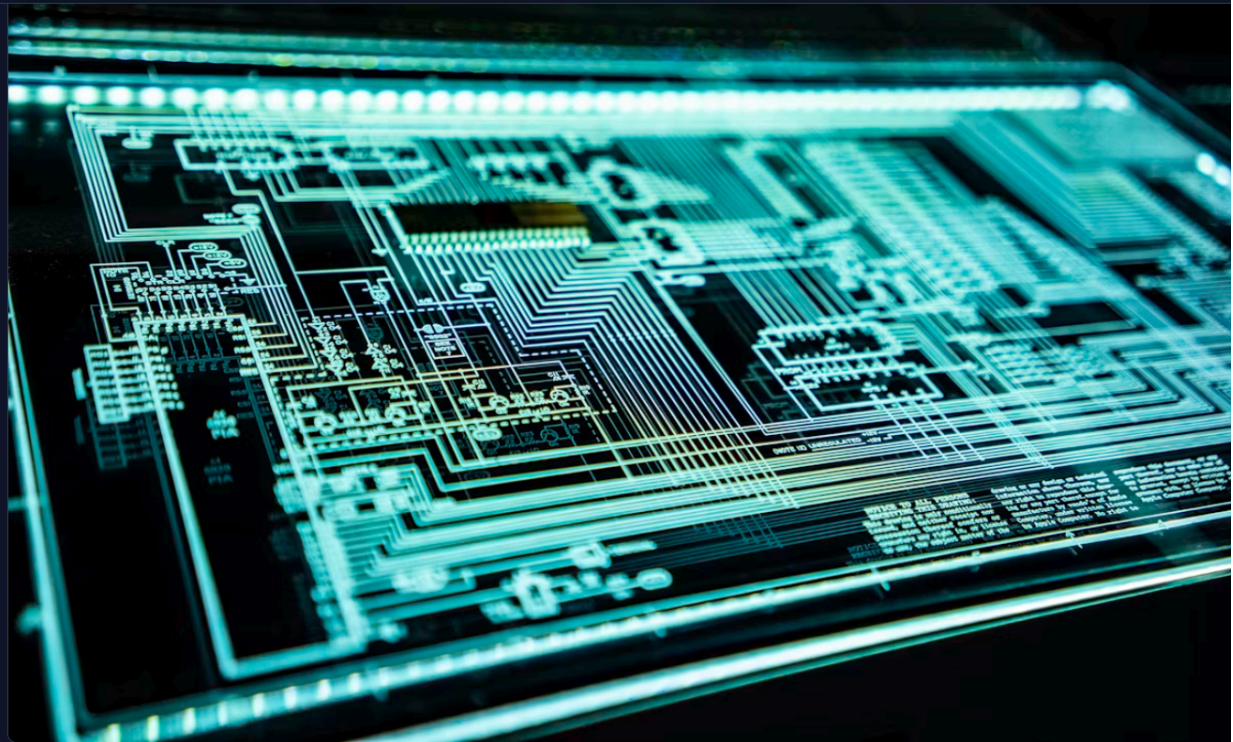
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Apex Dynamics

[Features](#) [Performance](#) [About](#)



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Limited understanding; misses key trends	Does not interpret or fails to understand data
Analysis	25%	Provides deep analysis with logical reasoning and justification	Provides reasonable analysis with some justification	Analysis with weak reasoning	Analysis is not meaningful or no analysis provided
Application of Concepts	20%	Accurately applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong evidence and implications	Draws valid conclusions with moderate insights	Conclusions are vague or weak	Does not draw or draws invalid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Clearly clear with minor issues	Lack of clarity or poor structure	Poor presentation and difficult to understand